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The Narrative Engine: A Genre-Agnostic Multi-Agent Architecture

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ABSTRACT:

Generative AI has completely changed how we create digital content, but the actual process is still pretty frustrating. If a user wants to create a story, turn it into a script, and generate matching concept art, they usually have to bounce between several different apps and keep tweaking their prompts. To fix this, we built a unified, multi-agent AI system that automates the whole process from just one single user prompt. We set up a web interface using Gradio and connected it to Large Language Models (LLMs) to create a step-by-step workflow. The system is split into four specific agents: a Story Agent to write the narrative, a Script Agent to format the dialogue, a Thematic Agent to pull out the deeper meaning, and a Visual Agent to generate the artwork. By linking these models together in one pipeline, our project proves that you can cut out the repetitive manual work and create a complete, multi-format creative package instantly.

Keywords: **Multi-Agent Systems, Generative AI, Large Language Models, Prompt Engineering, Web Integration**

1.Introduction:

Artificial Intelligence has made huge leaps recently, especially when it comes to generating text and images. Tools like Large Language Models (LLMs) and image generators are incredibly powerful. However, the way people actually use these tools to build larger projects is still very disjointed.

Right now, if you want to develop a complete creative project—like writing a story, adapting it into a screenplay, and creating concept art for it—you have to do a lot of manual work. You have to write a prompt for a text AI, copy the result, carry it over to another tool, write a new prompt to get a script, and then go to an image generator and try to describe the scene perfectly so the art matches the story. This involves a lot of trial and error just to keep the style consistent across everything.

To solve this problem, we developed "The Narrative Engine." This project introduces a multi-agent AI architecture designed to make the whole creative pipeline run smoothly in one place. Instead of relying on one massive prompt, we broke the job down and assigned different creative roles (like a Storyteller, a Screenwriter, an Analyst, and an Artist) to separate AI modules that work together in a sequence. Users just enter a basic idea into our Gradio web app, and the engine automatically hands back a fully formatted package of text and images.

Objective:

- To automate the generation of narrative and visual assets from a single textual prompt.
- To streamline the digital creative pipeline by unifying disparate AI generative tasks into a single framework.
- To eliminate the need for repetitive manual prompt engineering across multiple platforms.

2.Methodology:

The development of the "Narrative Engine" is grounded in a sequential, multi-agent AI framework. Early in our research, we identified that forcing a single Large Language Model (LLM) to simultaneously write a story, format a script, and generate visual prompts from one massive instruction often led to poor reasoning, AI hallucinations, and stylistic inconsistencies. To resolve this, our methodology shifts away from single-prompt generation and instead breaks the entire creative process down into discrete, specialized sub-tasks.

a. Technology Stack and System Integration For the core natural language processing and reasoning engine, we integrated Google's Gemini AI model[1]. Gemini was specifically chosen for this project because of its high-speed inference and strong ability to retain long context. For the visual generation component, the system routes data to a cloud-based Stable Diffusion model[2]. To tie these backend models together into an accessible prototype, we utilized Gradio[3], an open-source Python library, to build a lightweight, user-friendly web interface that anyone can operate without knowing how to code.

b. The Chained Data Flow Pipeline The system's logic relies on a linear, cascading data flow. The process begins at the user interface, where the user defines their initial parameters: a base textual idea (e.g., "A cybernetic detective investigating a neon city") and a target genre.

Instead of processing these inputs all at once, the system routes this "seed data" to our first AI module. Each agent in our system is programmed with a highly specific "system prompt" or persona so it knows exactly what its job is. Once the first agent generates the foundational narrative, the system automatically captures that text and injects it into the prompt for the second agent, and then the third.

c. Contextual Preservation By chaining the agents sequentially rather than running them in parallel, we ensure complete contextual preservation. The output of one model directly informs and constrains the next. For example, the visual AI isn't just guessing what the scene looks like based on the user's short initial prompt; it is generating art based on the highly detailed, descriptive narrative that the first agent just wrote. This chained methodology guarantees that the final screenplay, the thematic analysis, and the concept art are all perfectly aligned in tone, style, and subject matter, completely eliminating the need for the user to manually tweak prompts across different applications.

3.System Architecture:

. The architecture is broken down into three main sections: the user input area, the processing core where the AI agents live, and the final output dashboard.

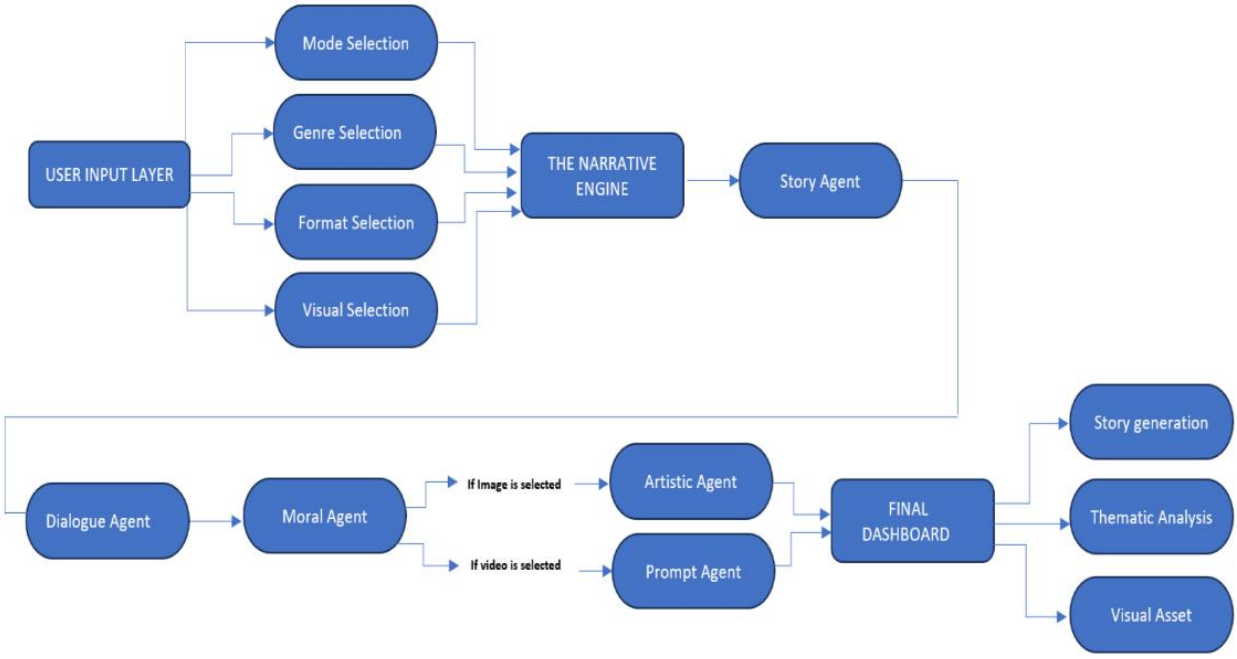


Fig:1.Flowchart

The processing core relies on four specific agents working together:

- **Agent 1 (The Narrative Storyteller):** This is the starting point. It takes the user's idea and genre and writes a short, creative story.
- **Agent 2 (The Screenwriter):** This agent waits for Agent 1 to finish. Once it gets the story, it rewrites the whole thing into a proper movie script or game quest, adding character dialogue and scene directions.
- **Agent 3 (The Thematic Analyst):** While the script is being written, this agent looks at the original story and figures out the core moral or philosophical theme behind it.
- **Agent 4 (The Visual Artist):** Finally, this agent takes the original idea and the genre and builds a highly detailed visual prompt. It sends this to a cloud-based image model (Stable Diffusion)[2] to generate high-quality concept art.
- The proposed multi-agent architecture was successfully deployed using a Gradio web interface. The prototype demonstrated the capability to ingest a single user prompt and distribute that context across multiple specialized AI modules.
- During testing, the system successfully routed the data sequentially: first to the Narrative Agent to establish the core story, and subsequently to the Script and Thematic Agents. The visual generation module simultaneously produced corresponding concept art. Table 1 demonstrates a successful test case of the sequential generation pipeline.

4.Results

Table 1: System Output for Sample User Prompt

User Input	An abandoned lighthouse in a foggy ocean
Selected Genre	Lovecraftian Horror
Selected Format	Movie Script
Agent 1 (Story)	The system successfully generated a 150-word foundational narrative establishing the dark, atmospheric tone of the lighthouse.
Agent 2 (Script):	The system converted the prose into a professionally formatted screenplay with scene headings (slug lines) and character dialogue.
Agent 3 (Moral):	The system extracted a core philosophical theme: Some beacons are not meant to guide, but to warn us away from the abyss
Agent 4 (Visual):	The Stable Diffusion routing successfully rendered an 8k resolution digital concept art piece matching the Lovecraftian aesthetic.

The results prove that chaining specialized AI agents significantly reduces the manual prompt engineering typically required to generate multi-format creative assets.

Conclusion

This paper has successfully presented "The Narrative Engine," a unified, multi-agent AI architecture designed to solve the growing problem of fragmented digital workflows. By developing a sequential pipeline that links Large Language Models (LLMs) with visual generation tools[4], we demonstrated that the complex process of multi-format content creation can be fully automated from a single user prompt.

During our testing, the prototype successfully distributed specific creative roles—such as narrative drafting, script formatting, thematic analysis, and visual concept generation—to specialized AI agents. This chained approach not only maintained strict stylistic and contextual consistency across all outputs but also eliminated the need for repetitive, manual prompt engineering. The integration of Google's Gemini for fast reasoning and Stable Diffusion for high-fidelity concept art, all wrapped within an accessible Gradio web interface, proves the technical feasibility of this end-to-end creative engine.

While the current iteration serves as a highly effective working prototype, the underlying framework establishes a strong foundation for a fully automated, production-level content studio. Future work will focus on migrating the system from a local testing environment to a dedicated cloud server to ensure continuous uptime and faster processing speeds. Furthermore, future iterations will expand the system's output capabilities to include AI-driven video generation, automated voice-over synthesis, and direct-to-PDF document formatting. Ultimately, this architecture demonstrates the immense potential for AI to evolve from a standalone, single-purpose tool into an integrated, collaborative creative team.

Reference

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